The neutral memory field in a fixed charge sector

Composite filtering, exact pinching and selected finite-time moments

Alan Day Chengheng*

8 November 2026 Local research draft

Abstract

We complete the central-sector transfer of the actual neutral memory component to local marked Wick descendants. A fixed-degree graph interpolation estimate shows that the covariance filter on the actual Jacobi correction agrees asymptotically with its Gaussian-chaos multiplier. Together with the preceding covariance tangent and uniform tails, this yields strong convergence on compact real-time intervals at every fixed finite Schatten order. At half filling, the previously established central source identity transfers the residual spectral argument and exact energy pinching. The resulting finite-time moments extend to controlled polynomial preparations and the single-selected functional, retaining its local-size boundary terms.

1 Model, object and scope

Use standard CAR and the overlapping complex Gaussian p-body Hamiltonian with the same static couplings in all sectors. Let $\mathcal{H}_m = \wedge^m \mathbb{C}^N$, $\|X\|_{r,m} = (\mathbb{E}\,\mathrm{tr}_m\,|X|^r)^{1/r}$ and $\mathrm{tr}_m = \binom{N}{m}^{-1}\mathrm{Tr}_{\mathcal{H}_m}$. Take $\lambda_N = p^2/N \to \lambda > 0$, fixed energy normalization, $\hbar = 1$, and string time t. Set

$$C_m I = \mathbb{E}H_m^2, \quad \Psi_m = \Phi_m/C_m, \quad \Phi_m(X) = \mathbb{E}_{H'} H'_m X H'_m,$$

where the independent copy only defines the deterministic covariance map: every actual observable still uses the original disorder. The exact neutral component is

$$B_{m,N}(t) = m\epsilon_m(1-\mathcal{P}_H)e^{-iH_m t/2}\Phi_m(e^{i\alpha_m H_m t})e^{-iH_m t/2}, \quad \alpha_m = 1 - p/m, \quad \epsilon_m = 1-\alpha_m^2. \quad (1)$$

We work on $|m - N/2| \le L_0 p$, so $m > p$ eventually. $\mathcal{P}_H X = \sum_e P_e X P_e$ includes whole degenerate energy blocks. Strong approximation below holds on this compact central window. The limiting selected-state statements are at $m = N/2$. The actual total memory field and its bilocal return still require their separate sector interfaces.

*AI research assistant (OpenAI).

2 The missing composite filter estimate

Let $\mathcal{N}$ count coupling Gaussian chaos and $\mathcal{R}$ be the output $U(N)$ rank. On the equivariant Gaussian-matrix Hilbert space,

$$\mathfrak{D} = p\mathcal{N} - \mathcal{R} \geq 0, \qquad \Psi_m = \mu_m(\mathcal{R}), \qquad \theta_{j,m} = \mu_m(jp), \quad \Theta_m = \theta_m(\mathcal{N}).$$

The positive-part extension is

$$\mu_m(x) = \prod_{a=0}^{p-1} \left(1 - \frac{x}{m-a}\right)_+ \left(1 - \frac{x}{N-m+p-a}\right)_+ .$$

On a broad central interval it is Lipschitz with $\mathrm{Lip}(\mu_m) \leq 12p/N$. These are the existing fixed-sector rank and covariance identities [3, 5]. Define the finite-parameter Jacobi polynomials and their actual error:

$$\begin{aligned} a_j &= \sum_{v<j} \theta_{v,m}, \quad Q_0 = 1, \quad Q_1 = x, \quad Q_{j+1} = xQ_j - C_m a_j Q_{j-1}, \\ Z_j &= \mathcal{W}_j(H^j), \quad E_j = Z_j - Q_j(H), \quad \eta_j = pE_j. \end{aligned} \tag{2}$$

Lemma 1. *For each fixed j, L_0,*

$$\|(\Psi_m - \Theta_m)\eta_j\|_{2,m} \leq 12\lambda_N (\|E_j\|_{2,m} \left\|\mathfrak{D}^2 E_j\right\|_{2,m})^{1/2} = O_{j,L_0}(p^{-1/2}).$$

Proof. The joint rank/chaos spectral decomposition gives $\|(\Psi_m - \Theta_m)X\|_2 \leq (12p/N)\,\|\mathfrak{D}X\|_2$. Spectral Cauchy–Schwarz gives $\|\mathfrak{D}E\|_2^2 \leq \|E\|_2 \left\|\mathfrak{D}^2 E\right\|_2$. Insert $\eta = pE$ and $p^2/N = \lambda_N$. The needed inputs, $\|E_j\|_2 = O(p^{-1})$ and $\left\|\mathfrak{D}^2 E_j\right\|_2 = O_j(1)$, are sectorwise; their audit follows. □

The first bound also follows directly from the preceding actual η_j envelope [7, 6]. For the second, the Racah sixth-moment certificate [4, 5] is polynomial in $a/\sqrt{N}, r/\sqrt{N}, (m - N/2)/\sqrt{N}$. Its denominators stay nonzero on compact sets, so it controls each fixed rank step with $a \leq p, r \leq jp$ on the present window. It bounds both the fourth deficit moment of the independent product chain and the fourth one-step moment.

Gaussian creation symmetrization contracts normalized sector L^2. The chain bound thus gives $\left\|\mathfrak{D}^2 Z_j\right\|_2 = O_j(1)$. For left multiplication $\mathcal{T} = H_{\mathrm{left}}$, the one-step bound gives bounded $[\mathfrak{D}, [\mathfrak{D}, \mathcal{T}]]$ on each fixed-degree space. The annihilation block is its adjoint. Iterating $\mathfrak{D}^2\mathcal{T} = \mathcal{T}\mathfrak{D}^2 + 2[\mathfrak{D}, \mathcal{T}]\mathfrak{D} + [\mathfrak{D}, [\mathfrak{D}, \mathcal{T}]]$ gives $\left\|\mathfrak{D}^2 H^n\right\|_2 = O_n(1)$, hence the same bound for $Q_j(H)$. No full-trace averaging is used.

3 From the tangent to every fixed neutral coefficient

On full neutral block families set

$$D = Q - \sum_i c_i^\dagger(\,\cdot\,)c_i, \qquad \bar{D} = N - Q - \sum_i c_i(\,\cdot\,)c_i^\dagger, \quad L = (D+\bar{D})/2, \quad S_v = (vp - L)Z_v.$$

Apply these maps before sector restriction. With target-sector scalars $C_m, \theta_{v,m}$ retained, the preceding tangent [8] is

$$A_v = p(\Phi_m - C_m\theta_{v,m})Z_v, \qquad \|A_v - 4\lambda_N C_m \theta_{v,m} S_v\|_{r,m} = O_{v,r,L_0}(p^{-1}). \tag{3}$$

The ordered coupling-Wick product is denoted by $\diamond$. Define the local forced recurrence

$$\begin{aligned} \widehat{r}_j^S &= 4\lambda_N C_m \sum_{v<j} \theta_{v,m}(S_v \diamond Z_{j-1-v}), \\ \widehat{\eta}_{j+1}^S &= H\widehat{\eta}_j^S - C_m a_j \widehat{\eta}_{j-1}^S - \widehat{r}_j^S, \qquad \widehat{\eta}_0^S = \widehat{\eta}_1^S = \widehat{\eta}_2^S = 0. \end{aligned} \tag{4}$$

The exact actual recurrence has the same form with forcing $\sum_{v<j} A_v \diamond Z_{j-1-v}$ [5, 6]. For a pure degree-v error X_v,

$$X_v \diamond Z_w = \mathcal{W}_{v+w}(X_v H^w), \qquad \|\mathcal{W}_d Y\|_r \le 2^d \|Y\|_r \quad (\deg_{\text{Gaussian}} Y \le d).$$

Fixed-degree Hölder and (3), followed by the finite recurrence, prove

$$\left\|\eta_j - \widehat{\eta}_j^S\right\|_{r,m} = O_{j,r,L_0}(p^{-1}) \quad (j, r \text{ fixed}). \tag{5}$$

This estimate does not assert a degree-uniform error constant.

The actual and local neutral coefficients are

$$\begin{aligned} T_j &= A_j + (C_m\theta_{j,m} - \Phi_m)\eta_j, \\ V_j^S &= 4\lambda_N \theta_{j,m} S_j + (\theta_{j,m} - \Theta_m)\widehat{\eta}_j^S. \end{aligned} \tag{6}$$

Their difference is exactly

$$\begin{aligned} C_m^{-1}T_j - V_j^S = {} & C_m^{-1}A_j - 4\lambda_N\theta_{j,m}S_j \\ & + (\theta_{j,m} - \Theta_m)(\eta_j - \widehat{\eta}_j^S) - (\Psi_m - \Theta_m)\eta_j. \end{aligned} \tag{7}$$

Here Θ_m is an L^2 contraction and C_m is bounded above and away from zero on the window. Lemma 1 and (5) give $O_j(p^{-1/2})$ in L^2. Both sides have bounded higher fixed Schatten norms. Interpolation therefore gives $\left\|T_j - C_m V_j^S\right\|_{r,m} \to 0$ for every fixed finite r. We do not claim the same rate at all r.

4 The actual infinite series, before energy pinching

The finite Jacobi coefficients satisfy the exact entire expansion and the uniform estimate [5, 6]

$$e^{itH} = \sum_{j\geq 0} c_{j,m}(t)Q_j(H), \qquad |c_{j,m}(t)| \leq e^{2\sqrt{AC_*}S}(2\sqrt{A}S)^j/j!, \quad |t| \leq S,$$

where $a_j \leq A$ and $C_m \leq C_*$. Removing the samplewise conserved function $\sum_j c_j\theta_j Q_j(H)$ gives the exact identity

$$p(1-\mathcal{P}_H)\Phi_m(e^{itH}) = (1-\mathcal{P}_H)\sum_{j\geq 0} c_{j,m}(t)T_j.$$

The actual A_j, η_j envelopes and CP contraction of Φ_m/C_m give, on the window and at fixed r,

$$\|T_j\|_{r,m} + \left\|V_j^S\right\|_{r,m} \leq K_r(j+1)^{b_r}L_r^j\sqrt{j!}. \tag{8}$$

For the local term this follows from the pure S_v envelope, the OU/Chebyshev projection bound and the fully expanded recurrence (4). Expand that recurrence before Hölder: its two product layers require only a fixed multiple of r, not an exponent doubling at every degree. The same argument controls $\Theta_m\widehat{\eta}_j^S$ by an exponential-in-j factor, absorbed in L_r. All these bounds are already sectorwise in [6, 7].

Define the finite local proxy

$$\widetilde{B}_{m,N}^S(t) = \frac{m\epsilon_m C_m}{p}(1-\mathcal{P}_H)e^{-iH_m t/2}\sum_{j\geq 0} c_{j,m}(\alpha_m t)V_j^S e^{-iH_m t/2}.$$

Theorem 1. *For every fixed finite r, S, L_0,*

$$\sup_{|m-N/2|\leq L_0 p}\sup_{|t|\leq S}\left\|B_{m,N}(t) - \widetilde{B}_{m,N}^S(t)\right\|_{r,m} \longrightarrow 0.$$

Proof. The tail of either series after J is bounded by a constant times

$$\sum_{j>J}(j+1)^{b_r}L_{r,S}^j/\sqrt{j!} \longrightarrow 0$$

uniformly in N, m, t. The external phases are unitary, $m\epsilon_m/p \leq 2$, and $\|1-\mathcal{P}_H\|_{r\to r} \leq 2$. At fixed J, apply (7) term by term. First take $N\to\infty$, then $J\to\infty$. □

The approximation uses the original common disorder and retains the finite-N energy projection. It uses the rank graph bounds only on fixed-degree errors; no high-Schatten bound on a nonlinear rank multiplier or rank-marked proxy is assumed.

5 Central limiting fields and exact removal of static mass

Now set $m = N/2$, $C_m \to C > 0$, $q = e^{-4\lambda} \in (0,1)$. Each term in (4) is a finite linear combination of

$$\mathcal{W}_d[H^a(S_v \diamond Z_w)].$$

Its L mark lies in a pure Wick subblock; any exterior H slots and overall chaos filter are allowed. The central all-G source identity of [10] therefore applies to arbitrary products of these primitives and their adjoints. Its local exponent agrees with the earlier tracial exponent, directly in the central sector. Thus the regression and two-energy proof of [9] transfers without inferring a sector law from a full-charge average.

Let $H_j^{(q,C)}$ be monic q-Hermite, $P_j(q) = [j]_q!$, and

$$\kappa_j = \lambda \frac{(j^2 - 4q\partial_q)P_j}{P_j}.$$

For $g_j = \operatorname{Reg} \widehat{\eta}_j^S$, the existing regression is

$$g_{j+1}(x) = xg_j(x) - C[j]_q g_{j-1}(x) - 4\lambda C \left(\sum_{v<j} q^v \kappa_v \right) H_{j-1}^{(q,C)}(x),$$
$$g_0 = g_1 = g_2 = 0, \qquad g_j = \sum_d g_{jd} H_d^{(q,C)}.$$

The regression of $V_j = 4\lambda q^j S_j + (q^j - q^{\mathcal{N}})\widehat{\eta}_j^S$ is

$$v_j(x) = 4\lambda q^j \kappa_j H_j^{(q,C)}(x) + \sum_{d \le j-2} (q^j - q^d) g_{jd} H_d^{(q,C)}(x).$$

Define $N_j = V_j - v_j(H)$. For finite N, use the same polynomial $v_{j,N}$ at (λ_N, q_N, C_m) and set $N_{j,N} = V_{j,N}^S - v_{j,N}(H_m)$.

In the two-energy cut-diagram expansion, subtracting v_j cancels all groups with no direct field bridge. If D bounds the two fields' raw length, the remaining normalized energy-Hermite coefficients obey

$$|M_{ab}| \le K_D (a+b+1)^D q^{\max(0,\min(a,b)-D)} \mathbf{1}_{|a-b|\le D}.$$

This square-summable bound gives an $L^2(\rho_{q,C} \otimes \rho_{q,C})$ spectral density, hence zero mass on the energy diagonal. It is the old fixed-degree proof [9], now with central moments.

The finite positive measure $\nu_N(A,B) = \mathbb{E}\,\mathrm{tr}_m[N_{j,N}^\dagger P_H(B) N_{j,N} P_H(A)]$ has this weak limit. Higher central moments give tightness and uniform integrability; the compact limiting moment problem identifies it. Portmanteau on the closed diagonal yields

$$\|\mathcal{P}_H N_{j,N}\|_{2,m}^2 = \nu_N(\{E = E'\}) \longrightarrow 0.$$

Higher fixed moments and pinching contraction upgrade this to all finite r. Since $v_{j,N}(H)$ commutes exactly with H, $(1 - \mathcal{P}_H)V_{j,N}^S - N_{j,N} = -\mathcal{P}_H N_{j,N} \to 0$.

6 Finite-time selected moments

The limiting neutral field is the convergent noncommutative L^r series

$$B_\infty(t) = 2Ce^{-iHt/2} \sum_{j\geq 1} c_j^{(q,C)}(t) N_j e^{-iHt/2}. \tag{9}$$

To justify every fixed finite r, first apply the preceding fixed-j pinching result. The norm of the limiting N_j is bounded by the limiting norm of $(1-\mathcal{P}_H)V_{j,N}^S$, hence by twice (8). This supplies the limiting series tail, without assuming a uniform high-norm bound on the entire regression map. The finite Jacobi coefficients converge on compact times; keep α_m and $m\epsilon_m/p$ until this fixed-truncation limit.

Hölder and telescoping transfer all fixed mixed covariant words in $B(t)$, its adjoint, fixed energy polynomials/phases, and the already controlled charged time fields. Times range over fixed compact sets. For a controlled fixed polynomial preparation $P_N O$,

$$\mathbb{E}\,\|X_N P_N O\|^2 \leq C_P \,\|X_N\|_{4,m}^2$$

by the finite-N orbit-density theorem [11]. Together with polarization this transfers errors inside $\mathbb{E}\langle O|P_N^\dagger \mathcal{B}_N Q_N|O\rangle$. Preparation degree stays fixed. Multiple arbitrary nonpolynomial selected factors interlaced with separate pinched blocks are not included merely by this preparation argument.

For one normalized selected excitation, put $h = H/\sqrt{C}$, $W = (S_1 - \lambda H)/\sqrt{C}$, $s = \sqrt{q}$. The full boundary functional [10] extends through the convergent tails:

$$\sigma_1(\mathcal{B}) = [1 - s(1+\lambda)]\tau(\mathcal{B}) + s\tau\big((h-W)\mathcal{B}(h-W)\big). \tag{10}$$

Use the actual phases in (9). This functional need not be stationary on the full marked algebra. The charged-only expression $\tau((1-s+sh^2)\mathcal{B})$ is not a replacement for (10) on neutral or mixed words.

As a calibration, $N_1 = 4\lambda q(S_1 - \lambda H) = 4\lambda q\sqrt{C}\,W$, $\tau(W^2) = \lambda$, and

$$\sigma_1(N_1^2) = 16\lambda^3 C q^2[1 + \lambda s(1+q)].$$

This recovers the earlier slope result in the complete-field interface. In string time,

$$\sigma_1(B_\infty'(0)^\dagger B_\infty'(0)) = 64\lambda^3 C^3 q^2[1 + \lambda\sqrt{q}(1+q)].$$

Using the charged-only density would omit the positive relative correction $\lambda\sqrt{q}(1+q)$. Rational source checks recover this difference exactly; they do not identify rational (γ, λ) with the physical curve $\gamma = e^{-\lambda}$.

7 Executed checks and the remaining return problem

The new finite checker keeps all Gaussian coordinates in $(N, p, m) = (3, 1, 2), (4, 1, 2), (4, 2, 2)$ through degrees $5, 4, 3$. An independently constructed CAR Casimir diagonalizes the rank filter; the covariance is also built directly from its Kraus matrices. Fifteen degree rows check the graph Cauchy–Schwarz inequality and the composite-filter estimate, with six nonzero filter rows. The largest direct-versus-spectral Hermite error is below $9.7 \cdot 10^{-16}$; these are floating-point consistency tests with explicit tolerances, not interval certificates.

The rational local-source compiler retains pure-Wick regions, local L subblocks, overall chaos and OU degree weights. At two independent rational (γ, λ) points it checks 16 two-energy moments against the old descendant evaluator and 40 regression cancellations. It also checks 16 selected boundary identities and two explicit counterchecks to the charged-only density. No finite-N energy pinching simulation is used as evidence for its limiting theorem.

The graph inputs are inherited, not recalculated: fixed-sector rank transition orthogonality, the compact-window sixth-moment certificate, and fixed-degree multiplication bounds. The decisive new combination is Lemma 1 applied to the *actual* pE_j, followed by (7). This avoids reconstructing a rank-proxy exponential just to resolve the composite filter.

All science outputs are preserved after their first successful run. The evidence manifest records code versions, parameters, inherited file hashes, the two-pass PDF and final rendered pages. No Git push or issue update accompanies this local stage.

Next physical interface. The actual neutral component is now controlled at fixed charge in the preparations just stated. The next task is to audit the finite-N total-memory decomposition into charged plus neutral components, including the collective return error and its conserved subtraction, then the Volterra return to the original bilocal. Sector-uniform force high moments alone do not prove these small-error statements. The full-trace bilocal calculation is already available and will be reused only where its proof is genuinely sectorwise. The full selected bilocal, spatial/mass bulk dictionary and finite-G dynamics remain unfinished.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983.

[2] E. G. Effros and M. Popa, *Feynman Diagrams and Wick Products Associated with q-Fock spaces*, arXiv:math/0303045. These are the fixed-charge and q-Wick combinatorial background, not substitutes for the present sector-error estimates.

[3] Project draft, `manuscript/susskind-memory-wick-rank/main.tex`, §§5–6: rank transition orthogonality and creation/annihilation bounds.

[4] Project draft, `manuscript/susskind-memory-central-window/main.tex`, §§2–3: the compact-window polynomial-ring certificate.

[5] Project draft, `manuscript/susskind-memory-neutral-marked-series/main.tex`, §§2–6: finite Jacobi recurrences and fixed-degree graph bounds.

[6] Project derivation, `notes/susskind-bilocal-fourth-tail-operator-draft.md`, §§2–7: actual CP residual, recurrence and neutral tails.

[7] Project draft, `manuscript/susskind-sector-gradient-tail/main.tex`: central-sector actual envelopes and charged time summation.

[8] Project draft, `manuscript/susskind-sector-covariance-tangent/main.tex`: the exact two-deficit remainder and fixed-degree local tangent.

[9] Project draft, `manuscript/susskind-memory-wick-descendants/main.tex`: ordered local sources, regression and the no-diagonal-mass proof.

[10] Project draft, `manuscript/susskind-slater-higher-wick/main.tex`: the all-G central source identity and selected boundary functional.

[11] Project draft, `manuscript/susskind-slater-polynomial-density/main.tex`: finite-N densities for fixed polynomial preparations.